Project Overview

Topic: Neurosymbolic Reasoning

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Contents

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Classical AI: Rule-based/Symbolic AI

Knowledge Base

$$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x) \qquad \text{(All humans are wibu.)}$$

$$\text{Human}(\text{Huy}) \qquad \text{(Huy is human.)}$$

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$$\begin{array}{c} A \rightarrow B \\ A \\ \hline \therefore B \end{array}$$

(Syllogism/Modus ponens)

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Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

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- Scalability
- Poor Perception

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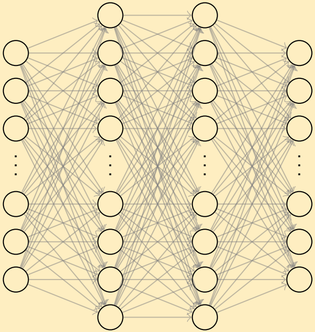
$\Rightarrow$ **reliable but inflexible.**

Modern AI: Data-driven/Neural Networks

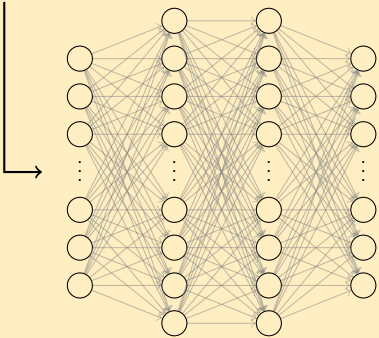
Neural Networks



Modern AI: Data-driven/Neural Networks

Neural Networks

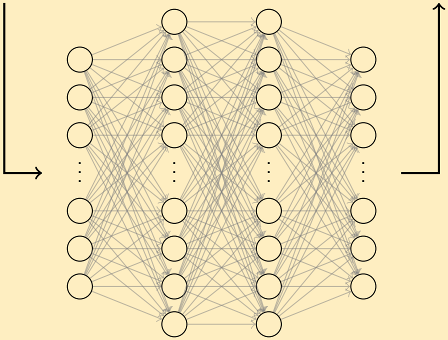
"Is Huy a good boy?"



Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy a good boy?” “No, he’s wibu.”



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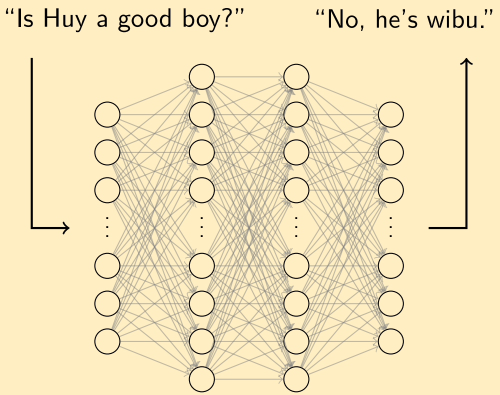

Pros

- Generalization & Robustness
- Perception
- Automated Feature Extraction

Modern AI: Data-driven/Neural Networks

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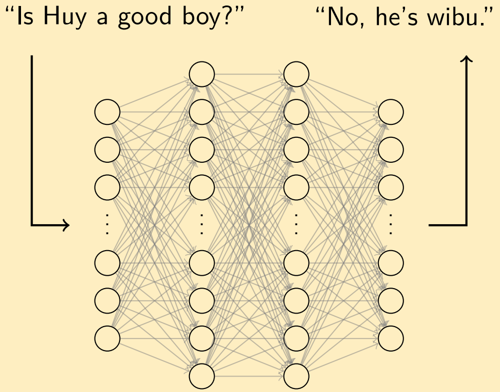

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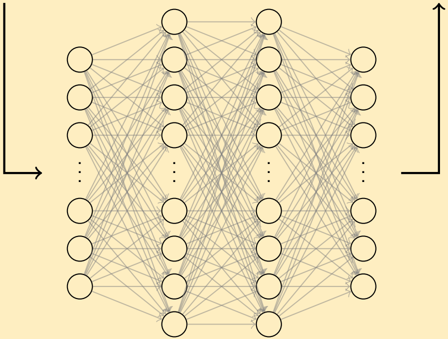

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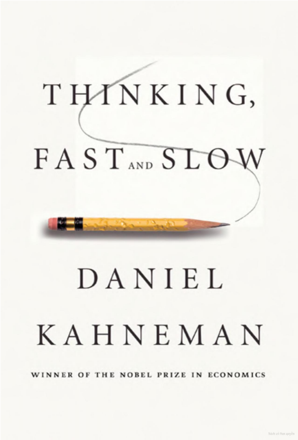
$\Rightarrow$ **flexible but unreliable.**

Neurosymbolic AI: A Hybrid Approach



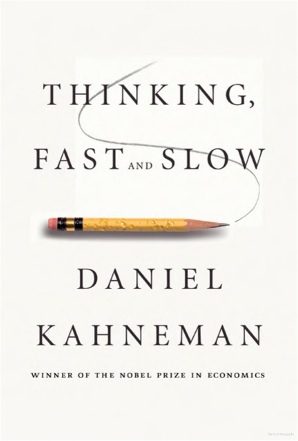
SYSTEM 1 (FAST)

Intuition & Instinct

SYSTEM 2 (SLOW)

Rational thinking

Neurosymbolic AI: A Hybrid Approach



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NEURAL NETWORK

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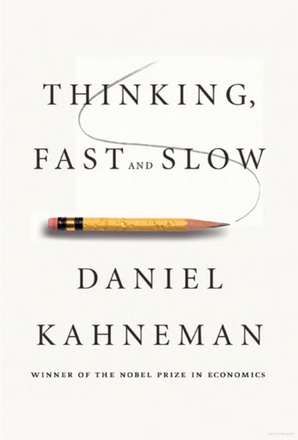

SYSTEM 1 (FAST)	**SYSTEM 2 (SLOW)**
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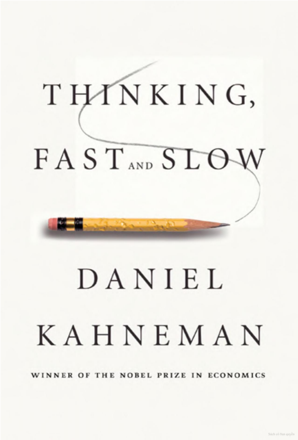
‖	‖
NEURAL NETWORK	**SYMBOLIC ENGINE**

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)
Intuition & Instinct

+

SYSTEM 2 (SLOW)
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=

HUMAN INTELLIGENGE

||

NEURAL NETWORK

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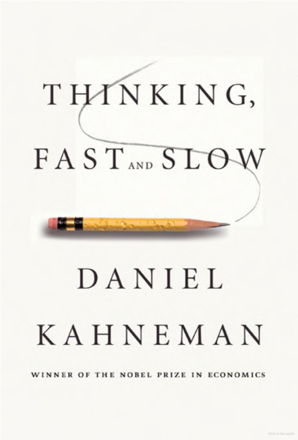
Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST) (Intuition & Instinct) + **SYSTEM 2 (SLOW)** (Rational thinking) = **HUMAN INTELLIGENGE**

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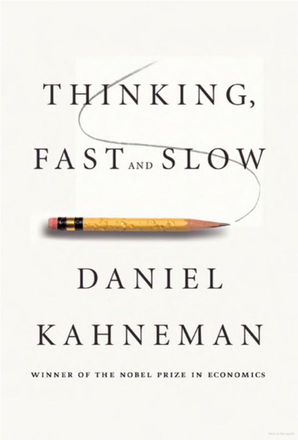
NEURAL NETWORK + **SYMBOLIC ENGINE** = **NEUROSYMBOLIC AI**

Neurosymbolic AI: A Hybrid Approach



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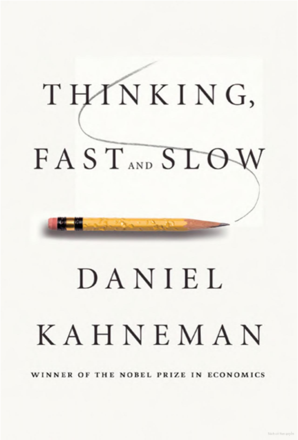
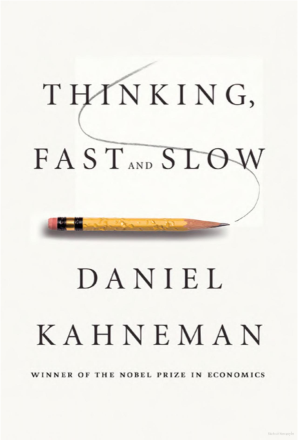


NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

Neurosymbolic AI: A Hybrid Approach



Semantic Parsing/Language Formalization

↑

NEURAL NETWORK ⊕ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

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Problem Statement

Language Formalization

The goal is to learn a mapping function $f : \mathcal{X} \to \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

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Input (x_{NL})

Natural Language
(Unstructured)

"Every student takes a course."

$$\xrightarrow{f(\cdot)}$$

Output (y_{FOL})

First-Order Logic
(Structured)

$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \wedge \text{Takes}(x, y)))$

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Core Challenges:

- **The Requirement for Absolute Precision:** Formal languages are syntactically fragile. A formal expression must be 100% correct to be machine-verifiable.
- **Data Scarcity and the Expert-Data Bottleneck:** Creating NL-FOL pairs requires professionals with deep knowledge of both the domain and the formal syntax.

Contents

Data Source

The data used consists of the **training sets** of the datasets listed in the table below.

Source	**Volume**	**Description**
FOLIO	1,000 samples	**With FOL label**, with true/false/uncertain label.
ProofWriter	41,400 samples	**Without FOL label**, with true/false label, with proof.
RuleTaker	480,000 samples	**Without FOL label**, with entailment/not entailment label.
ReClor	4,637 samples	**Without FOL label**, in multiple-choice format.
LogiQA	7,380 samples	**Without FOL label**, in multiple choice format.

Table: Preliminary statistics of data sources.

Data Source

FOLIO Dataset

FOLIO Dataset

FOLIO is an expert-written, open-domain, logically complex and diverse dataset for natural language reasoning with first-order logic.

Examples:

Premises	Premises-FOL	Conclusion	Conclusion-FOL	Label
All rabbits have fur. Some pets are rabbits.	$\forall x\ (\mathtt{Rabbit}(x) \rightarrow \mathtt{Have}(x, \mathtt{fur}))$ $\exists x\ (\mathtt{Pet}(x) \land \mathtt{Rabbit}(x))$	Some pets do not have fur.	$\exists x\ \exists y\ (\mathtt{Pet}(x) \land \mathtt{Pet}(y) \land \neg\mathtt{Have}(x, \mathtt{fur}) \land \neg\mathtt{Have}(y, \mathtt{fur}))$	Uncertain
All tigers are cats. No cats are dogs. All Bengal tigers are tigers. All huskies are dogs. Fido is either a Bengal tiger or a cat.	$\forall x\ (\mathtt{Tiger}(x) \rightarrow \mathtt{Cat}(x))$ $\forall x\ (\mathtt{Cat}(x) \rightarrow \neg\mathtt{Dog}(x))$ $\forall x\ (\mathtt{BengalTiger}(x) \rightarrow \mathtt{Tiger}(x))$ $\forall x\ (\mathtt{Husky}(x) \rightarrow \mathtt{Dog}(x))$ $\mathtt{BengalTiger}(\mathtt{fido}) \oplus \mathtt{Cat}(\mathtt{fido})$	Fido is neither a dog nor a husky.	$\neg\mathtt{Dog}(\mathtt{fido}) \land \neg\mathtt{Husky}(\mathtt{fido})$	True
...	...	...	...	...

Table: Sample data from the FOLIO dataset.

Data Source

ProofWriter Dataset

ProofWriter Dataset

ProofWriter is ...

Examples:

Data Source

RuleTaker Dataset

RuleTaker Dataset

RuleTaker is ...

Examples:

Data Source

ReClor Dataset

ReClor Dataset

ReClor is ...

Examples:

Data Source

LogiQA Dataset

LogiQA Dataset

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Examples:

Data Pipeline

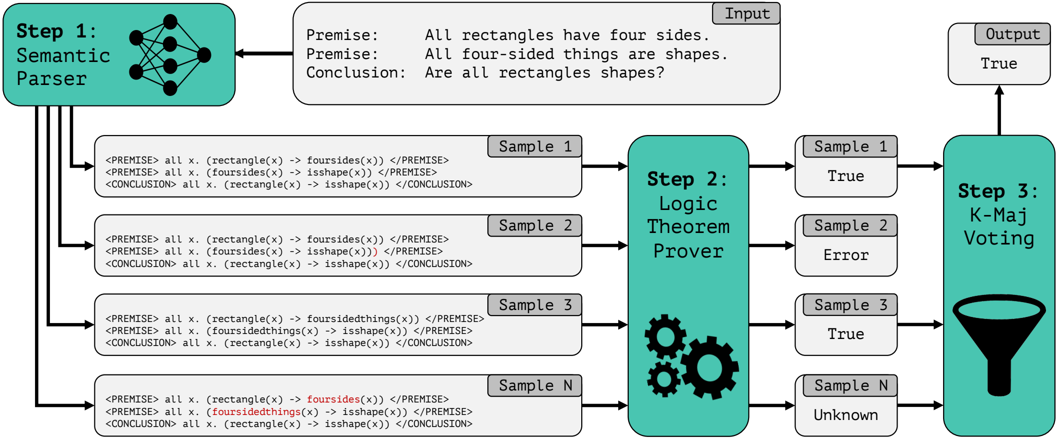

Figure: Mình không hẳn là dùng pipeline này, để tạm đây mai sửa

Data Pipeline

(Viết chi tiết aabcxyz về data pipeline, giai đoạn 1 dùng SFT, giai đoạn 2 dùng DPO)

Supervised Fine-Tuning (SFT):

$$L_{\mathrm{SFT}}(\theta) = -\sum \log \mathbb{P}(y_{\mathrm{FOL}} \mid x_{\mathrm{NL}}).$$

Direct Preference Optimization (DPO):

$$L_{\mathrm{DPO}}(\pi_\theta; \pi_{\mathrm{ref}}) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\mathrm{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\mathrm{ref}}(y_l|x)} \right) \right].$$

Contents

Related Work

Liệt kê ngắn gọn đóng góp của các paper liên quan

Contents

Anticipated Challenges & Proposed Solutions

Challenge X

XXX

Solution Y

YYY

Thank you!